

Trinh Hai Tien

Hanoi, Vietnam

Phone +84 879290105

E-mail trinhhaitienemail@gmail.com

LinkedIn linkedin.com/in/tien-trinh

GitHub github.com/sieuanhta

Location Hanoi, Vietnam

Computer science undergraduate interested in mathematical and computational approaches to decision-making under uncertainty, with current research in mechanism design, statistical learning, and quantitative finance.

Education

Vietnam National University, Hanoi – University of Engineering and Technology (VNU-UET)

2023 – 2027

Bachelor of Science in Computer Science

- GPA: 3.60/4.00
- Dissertation: “The Price of Learning Quality in Finite-Inventory Information Procurement”
- Advisor: Dr. Duyen Thi Ngo

Publications

Manuscripts Under Review

- [5] **Trinh, H. T.** *Learning and Incentives in Finite-Inventory Data Procurement*. Under review.
Fields: Mechanism Design, Statistical Learning, Information Economics
- [4] **Trinh, H. T.** *Maintaining Stable Coalitions under Sequential Agent Arrivals*. Under review.
Fields: Multi-Agent Systems, Hedonic Games, Online Algorithms

Manuscripts in Preparation

- [3] **Trinh, H. T.** *Derivative-Faithful Neural Networks*. Manuscript in preparation.
Fields: Scientific Machine Learning, Surrogate Modeling, Derivative Approximation

Peer-Reviewed Conference Proceedings

- [2] **Trinh, H. T.**, Man, T. B. P., Khuong, V. T. A., Nguyen, L. T., Le, T. H., & Ngo, T. D. (2025). Enhancing micro-expression recognition via multi-task learning with demographic-aware auxiliary supervision. *Proceedings of the International Conference on Knowledge and Systems Engineering (KSE 2025)*.
- [1] **Trinh, H. T.**, Nguyen, B. D., Nguyen, L. T., Le, T. H., & Ngo, T. D. (2026). Disentangling identity and motion information in micro-expression videos for micro-expression recognition. *Proceedings of the Conference on Information Technology and its Applications (CITA 2026)*.

Presentations

Conference Presentations

- [2] **Trinh, H. T.** (2025). *Enhancing Micro-Expression Recognition via Multi-Task Learning with Demographic-Aware Auxiliary Supervision*. Paper presented at the *International Conference on Knowledge and Systems Engineering (KSE 2025)*.
- [1] **Trinh, H. T.** (2026). *Disentangling Identity and Motion Information in Micro-Expression Videos for Micro-Expression Recognition*. Paper presented at the *Conference on Information Technology and its Applications (CITA 2026)*.

Research Experience

Research Assistant

2024 – Present

Human–Machine Interaction Laboratory | VNU University of Engineering and Technology

Supervisor: Dr. Ngo Thi Duyen

- Conduct research across machine learning, mechanism design, statistical inference, and computational decision-making.
- Study finite-inventory information procurement under strategic behavior and unknown data quality, developing truthful procurement mechanisms, randomized auditing schemes, and theoretical guarantees on the cost of learning quality.
- Investigate population-informed neural surrogate models for recovering first- and second-order differential information from sparse value-only observations, with emphasis on differential identifiability and derivative accuracy.
- Earlier research on subject-independent micro-expression recognition explored demographic-aware multi-task learning and identity–motion representations, resulting in two peer-reviewed conference papers at KSE 2025 and CITA 2026.

Researcher

2026 – Present

Beetrade

- Conduct research on quantitative trading strategies using historical financial-market data, with emphasis on statistical validation and market microstructure.
- Develop and evaluate medium-frequency strategies involving signal construction, backtesting, portfolio construction, and execution.
- Investigate alpha generation, statistical arbitrage, execution optimization, and multi-agent approaches to trading-system design.

Honors and Awards

Academic & Research Honors

- **Bronze Award**, Student Scientific Research Competition, VNU-UET 2026
- **Silver Award**, Student Scientific Research Competition, VNU-UET 2025
- **Robotics Challenge Fellowship**, Japan 2024
- **Bronze Medal**, Vietnam National Mathematics Olympiad 2023
- **Silver Medal**, VNU University of Science Mathematics Competition 2023
- **Bronze Medal (Mathematics)**, Vietnam Coastal and Northern Delta Science and Humanities Competition 2023

Selected Competitions

- **7th Worldwide (of 18,803 teams; 1st in Asia)**, IMC Prosperity 4 Algorithmic Trading Competition 2026
- **Silver Medal**, Vietnam National University Chess Tournament 2025

Scholarships

- **Academic Encouragement Scholarship**, VNU University of Engineering and Technology 2026
- **Gifted Student Scholarship** 2020 – 2023

Skills

Programming & Scientific Computing: Python, C++ , Java, Rust, Mathematica, LaTeX

Machine Learning & Data: PyTorch, DeepXDE, NumPy, pandas, scikit-learn; deep learning, automatic differentiation, numerical optimization

Mathematical & Quantitative Methods: probability, statistics, optimization, game theory, mechanism design, time-series analysis, numerical methods

Research & Development Tools: Git, GitHub, Jupyter Notebook, VS Code, Conda

Languages

Vietnamese: Native

English: Professional proficiency; IELTS Academic 7.5 (June 2026)